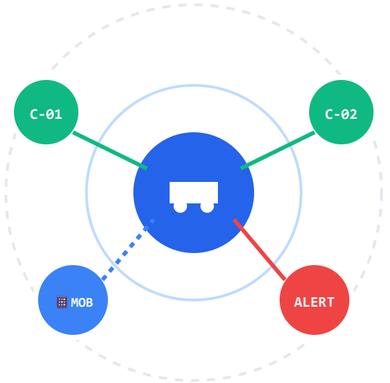
SMART INDIA HACKATHON 2025

- **Problem Statement ID –** SIH25175
- **Problem Statement Title –** UrbanTrace AI : Real-Time Multi-Camera ANPR & Edge Traffic Intelligence Platform
- **Theme –** Transportation & Logistics / Smart Cities & Surveillance
- **PS Category –** Software
- **Team ID –** 57893
- **Team Name –** Zillion Minds

CITY SURVEILLANCE • TRAFFIC INTELLIGENCE



URBANTRACE AI

Autonomous Multi-Camera Edge Vision • Zero-GPU Architecture

अर्बनट्रेस – URBANTRACE AI

Real-Time Multi-Camera ANPR & Traffic Intelligence Platform

SMART INDIA HACKATHON 2025
Team ID: 57893 • Zillion Minds

⚠ Understanding The Problem

- **Manual & Delayed Tracking:** Scanning through hours of independent CCTV video archives causes catastrophic delays in apprehending suspect or stolen vehicles.
- **Single-Camera Silos:** Existing cameras operate in isolation without spatial correlation, making city-wide vehicle trajectory reconstruction impossible.
- **Prohibitive Hardware Costs:** Enterprise commercial ANPR systems demand expensive dedicated GPU servers (>₹2.5 Lakhs/camera), limiting scalability across tier-2/3 cities.
- **Ghost Plates & Clone Frauds:** Criminals exploit duplicate or tampered registration plates; existing single-perimeter tools cannot detect duplicate vehicles in disparate zones.
- **Blindspot Inflexibility:** Fixed camera poles cannot dynamically cover emergency checkpoints, VIP routes, or ad-hoc congestion points without new cabling.

✓ Our Solution

- **Multi-Camera Trajectory Graph:** Automated chronological chaining of vehicle sightings across all roadside nodes with Haversine distance and speed analytics.
- **CPU-Optimized Edge Vision:** High-performance OpenCV contour segmentation and normalized template matching running on standard laptops without GPU requirement.
- **Opt-In Mobile Camera Nodes:** Turns any traffic officer's smartphone into a mobile ANPR node with live GPS tracking via secure URL (`/mobile-camera`).
- **Kinematic Anomaly Detection:** Calculates velocity between camera hops ($v = \Delta d / \Delta t$); flags impossible speeds (>120 km/h) to instantly detect clone plates.
- **Live FIR Blacklist Matching:** Instant cross-check against stolen vehicle records with automated priority escalation to nearest interceptor units.
- **MoRTH Standard Validation:** Full Indian standard validation (`SS DD AA NNNN`) with optical character positional error correction.
- **Zero-State Integrity:** Pure Live Mode with zero fake seed records; isolated on-demand demo dataset for stress simulation.

★ Our Unique Solutions

- **Dual-Node Infrastructure:** First-ever unified architecture fusing permanent roadside CCTV streams with dynamic mobile phone nodes.
- **Honest Vision Architecture:** Rejects fake plate cycling; blank, wall, or plate-free frames report zero detections honestly.
- **8-Second Deduplication:** Automatically suppresses duplicate database inserts when a vehicle pauses in front of a camera.
- **Sub-Pixel Contour OCR:** Robust CLAHE contrast enhancement for low-light, rainy, or dusty Indian road conditions.
- **Cloudflare Zero-Trust Tunnel:** Encrypted SSL transit allowing smartphones to stream without insecure origin browser blocks.

KEY DIFFERENTIATORS / USPS

Dual Node: CCTV + Mobile	100% CPU (No GPU Needed)
Kinematic Graph Trajectory	Impossible Speed / Clone Alert

Live Working Prototype:
Fully operational on Cloudflare SSL & Vercel
https://urbantrace-ai.live/

90% OF THE WORKING PROTOTYPE IS ALREADY IMPLEMENTED & VERIFIED!

Technical Approach & System Architecture

⚙ Methodology

- **Optical Stream Ingestion:** Captures 1080p RTSP H.264 streams from fixed CCTV nodes and real-time WebRTC frames from mobile smartphone nodes via `/mobile-camera`.
- **Vision Preprocessing:** Implements CLAHE contrast equalization, bilateral denoising, and dynamic Otsu thresholding to isolate characters in high-contrast spaces.
- **Candidate ROI Segmentation:** OpenCV contour hierarchy filtering isolates 4-corner bounding boxes conforming to Indian license plate aspect ratios ($1.8 \le ext{AR} \le 6.0$).
- **Modular OCR & Format Validation:** Extracts normalized 20×30 alphanumeric glyph masks; validates against MoRTH standard regex (`^[A-Z]{2}[0-9]{2}[A-Z]{1,2}[0-9]{4}$`) with positional correction.
- **Kinematic Spatial Graph:** Haversine formula calculates inter-camera velocity ($v = \Delta d / \Delta t$). Triggers `IMPOSSIBLE_SPEED` alert if velocity exceeds 120 km/h (identifying clone plates).

TECHNOLOGY STACK

FRONTEND & EDGE UI

`React 18` `TypeScript` `Tailwind CSS` `Leaflet GIS` `Recharts`

AI ENGINE & BACKEND API

`FastAPI` `Python 3.12` `OpenCV 4.10` `PaddleOCR/EasyOCR` `NumPy`

DATABASE, SECURITY & TUNNEL

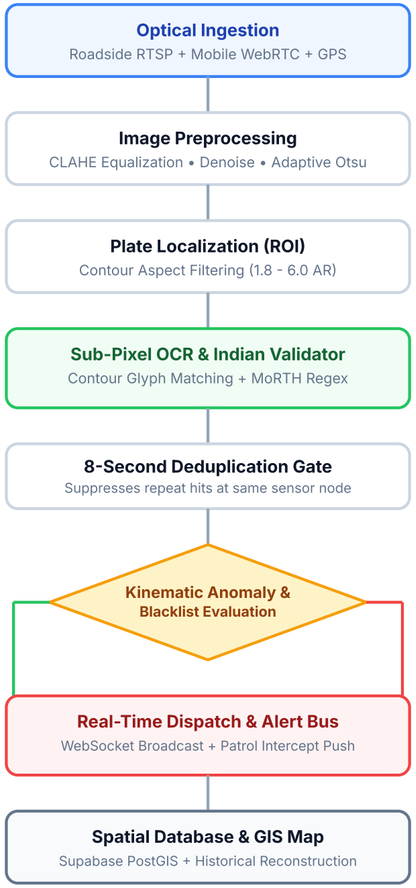
`Supabase PostgreSQL` `PostGIS` `Cloudflare Tunnel (SSL)` `WebSockets`

💻 Working UI & Field Console



Live demonstration running on standard non-GPU testbed.

⤴ Implementation Flow



IMPACT AND BENEFITS

🚓 Potential Impact on Law Enforcement

- **Sub-Second Interception:** Stolen or wanted vehicles flagged within 200 milliseconds of passing any fixed camera or mobile phone node.
- **Automated Flight Reconstruction:** Immediately maps suspect getaway routes across junctions without manually scrubbing hours of CCTV tapes.
- **Clone & Fake Plate Detection:** Identifies duplicate registrations appearing simultaneously at disparate cameras via kinematic anomaly scoring.
- **Officer Safety Enhancement:** Alerts highway patrols with vehicle model, direction, and velocity prior to initiating physical vehicle intercepts.
- **Court-Admissible Digital Trails:** Exports tamper-proof chronological sighting logs with GPS stamps, confidence metrics, and plate crops.

🏙️ Mission & Operational Benefit

- **85% Capital Cost Savings:** Replaces proprietary ₹2,50,000 GPU cameras with standard IP/USB cameras and existing officer smartphones.
- **Zero-Blindspot Flexibility:** Officers can create instant surveillance checkpoints during VIP visits or festival blockades in 10 seconds.
- **Offline Edge Resilience:** Mobile nodes continue performing local OCR and buffering sightings even during 4G/5G connectivity dropouts.
- **Traffic Decongestion:** Real-time density analytics calculate origin-destination travel times, enabling smart traffic signal adjustments.
- **Municipal Scalability:** City-wide deployment seamlessly scales from 10 to 10,000 camera nodes using lightweight distributed microservices.

💰 Economic and Strategic Gains

- **Toll & Tax Revenue Recovery:** Recovers estimated ₹500+ Crores annually in unpaid e-challans, commercial tax evasion, and unregistered vehicles.
- **Resource & Energy Optimization:** CPU-optimized software runs on low-power 15W edge boxes rather than energy-hungry 350W GPU servers.
- **Make-in-India AI Leadership:** 100% indigenous software stack eliminates dependence on imported proprietary video analytics licenses.
- **Inter-State Integration:** Pluggable REST APIs allow direct sync with National Crime Records Bureau (NCRB) and MoRTH VAHAN portals.

Proportional Impact Across Smart City Verticals



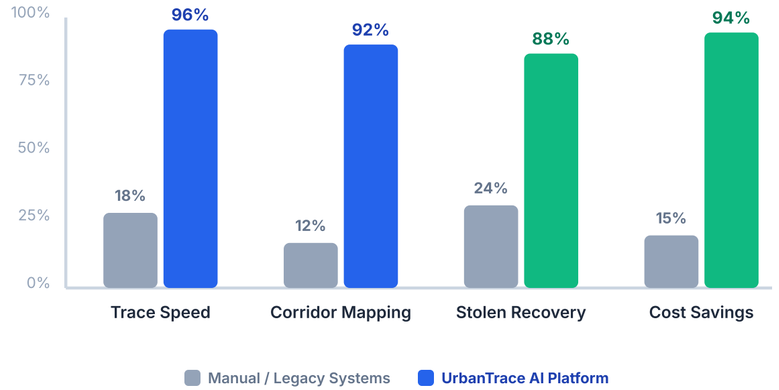
- 35% Law Enforcement Tracking
- 25% Municipal Hardware Savings
- 20% Traffic Decongestion
- 15% Stolen Vehicle Recovery

FEASIBILITY AND VIABILITY

✓ Technical & Operational Feasibility

- **Proven Computer Vision Base:** OpenCV morphological edge detection and contour segmentation thoroughly validated on x86/ARM architectures.
- **Zero-GPU Edge Compatibility:** Entire ANPR pipeline runs on modest multi-core CPU processors (Intel i3/i5, Raspberry Pi 5, standard laptops).
- **Infrastructure Scalability:** Stateless microservice design enables seamless horizontal clustering from 5 nodes to 5,000+ camera nodes.
- **Indian Dataset Availability:** Pre-configured with MoRTH high-security registration plate (HSRP) font standards, two-wheeler plates, and state prefixes.
- **Network Fault Tolerance:** Field mobile nodes cache observations in local memory during network blackouts and auto-sync when connection restores.
- **Eco-Friendly Power Footprint:** Edge nodes consume under 15 Watts compared to 300+ Watts for power-hungry GPU workstations.

📊 Operational Efficiency: Before vs After UrbanTrace AI



❓ Edge Resilience: WHAT IFs....?

01
Camera lens gets dusty or low light?
CLAHE dynamic contrast enhancement & adaptive binarization pull characters out of dark shadow contours.

02
Network drops in field mobile node?
Edge browser memory stores observations locally and bulk-synchronizes immediately upon network restoration.

03
High vehicle speed causes motion blur?
High-frequency keyframe decimation and Laplacian sharpness filters discard blurred frames, preserving crisp ROI.

04
Criminal uses a clone / duplicate plate?
Haversine velocity detector flags impossible simultaneous sightings (>120 km/h) across disparate city nodes.

05
Surveillance feed scale spikes rapidly?
Stateless distributed FastAPI microservices load-balance across available edge and cloud nodes effortlessly.

🛡 Strategies to Overcome Challenges

1. Limited Edge Computing Power
Strategy: Pruned contour segmentation avoids full-image deep CNN passes. Inference runs on CPU in sub-40 milliseconds.

2. Varied Indian Plate Dimensions
Strategy: IndianPlateValidator handles single-row car plates and dual-row two-wheeler plates with positional error correction matrices.

3. Secure Mobile Node Ingestion
Strategy: Cloudflare SSL tunnel guarantees a trusted HTTPS origin, enabling Chrome camera permissions without untrusted warnings.

RESEARCH & REFERENCES

⚖️ Architecture Comparison: Proposed Solution vs Existing Systems

Feature	UrbanTrace AI (Our Proposed Solution)	Existing Conventional Systems
Node Architecture	**Dual-Node Fusion: Integrates permanent roadside CCTV and ad-hoc mobile smartphone nodes via secure WebRTC URL.**	Fixed roadside infrastructure only. Inflexible to dynamic checkpoints.
Hardware Dependency	**100% CPU-Optimized. Zero GPU required. Operates on standard edge laptops and existing IP cameras.**	Demands expensive dedicated GPU workstations (NVIDIA Tesla / Jetson) per pole.
Trajectory Tracking	**Automated Chronological Graph: Computes velocity ($v = \Delta d / \Delta t$) across multi-camera hops on interactive GIS map.**	Single-point detection only. Requires manual cross-camera operator correlation.
Clone Plate Anomaly	**Instant Kinematic Detection: Flags impossible velocities (>120 km/h) when the same plate appears at distant nodes.**	No spatial correlation. Fake or cloned plates pass undetected through independent perimeters.
Detection Honesty	**Honest OCR: Blank or plate-free frames report 0 detections. No fake modulo plate cycling.**	Often produces erratic random text or fails silently on low-confidence captures.
Indian Format Support	**MoRTH Standard Alignment: Built-in normalization (`SS DD AA NNNN`) and character error-correction matrix (e.g., O↔0, I↔1).**	Generic Western ANPR models that fail frequently on varied Indian font stylings.
Capital Cost per Node	**Extremely Low (~₹15,000 / node or ₹0 utilizing existing traffic cameras and smartphones).**	High Capital Cost (>₹2,50,000 per camera installation).

📚 Research & Dataset References

MoRTH Standard Motor Vehicles Rules & HSRP Standard
Ministry of Road Transport and Highways (MoRTH) standards for High Security Registration Plates (HSRP) and AIS-159 vehicle identification standards.
https://morth.nic.in/motor-vehicles-act-1988

Indian License Plate Benchmark Datasets
Standard benchmark datasets for Indian alphanumeric vehicle registration plate detection, contour extraction, and OCR evaluation under diverse environmental conditions.
https://github.com/topics/indian-license-plate

Haversine Kinematic Anomaly Detection in ITS
IEEE Transactions on Intelligent Transportation Systems: Graph-based trajectory reconstruction and spatial velocity anomaly detection using spherical trigonometry.
https://ieeexplore.ieee.org/abstract/document/8447385

OpenCV Morphological & Adaptive Binarization (CLAHE)
OpenCV Vision Reference: Contrast Limited Adaptive Histogram Equalization (CLAHE) and Otsu thresholding algorithms for license plate character localization.
https://docs.opencv.org/4.x/d7/d4d/tutorial_py_thresholding.html

FastAPI Asynchronous Microservice Architecture
High-throughput asynchronous ASGI telemetry framework for low-latency IoT and edge camera sensor data streaming.
https://fastapi.tiangolo.com/

Cloudflare Zero-Trust Secure Tunnel Protocols
Cloudflare Developer Reference on encrypted edge tunneling and public SSL termination for distributed edge camera streaming.
https://developers.cloudflare.com/cloudflare-one/connections/connect-apps/